Selection (or conditional) logic in C#

Programs do not have to run every line of code in them. There can be decision points in which the program branches and follows a different path through the program. These decision points or conditional statements can either be IF-ELSE statements or SWITCH statements.

IF-ELSE statement

The IF statement is the most widely used statement in programming. The compiler evaluates an expression and if it is true then it executes the IF block of code, otherwise it executes the ELSE block of code. The ELSE block is optional. A block of code usually starts with an opening curly bracket { and ends with a closing curly bracket }. The brackets are optional if there is only one statement in a block.

```
int number;
Console.WriteLine("Please enter a number less than 10");
number = int.Parse(Console.ReadLine());
if(number < 10)
        Console.WriteLine("Good job!");
//Console.ReadLine();
```

In the example above, Console.ReadLine() reads in whatever the user inputs into a line on the console window. It puts that value into a variable called "number". Then it evaluates the expression between the round brackets. If the expression evaluates to true, then it displays the message, otherwise it does not.

Here is the same example, but this time with an ELSE clause which will run if the expression evaluates to false.

```
Console.WriteLine("Please enter a number less than 10");
int number = int.Parse(Console.ReadLine());
if(number >= 10)
        Console.WriteLine("The number should be < 10");
else {
    Console.WriteLine("Yes - the number is < 10");
    Console.WriteLine("Good job!");
}
```

It is possible to combine more than one IF statement together. The example below shows a nested IF statement.

```
Console.WriteLine("Please enter a number between 0 and 10");
   int number = int.Parse(Console.ReadLine());
   if(number > 10)
           Console.WriteLine("The number should be <= 10");
       else
           if(number < 0)
               Console.WriteLine("The number should be >= 0");
           else
               Console.WriteLine("Good job!");
```

Ternary If statement

There is also a shorthand notation for writing the If-Else statement all on one line. It has the following format:

(condition) ? ExpressionIfTrue : ExpressionIfFalse

For example,

```
Console.WriteLine("Please enter a number");
int number = int.Parse(Console.ReadLine());
if (number % 2 == 0)
{
      Console.WriteLine("The number is even");
}
else
{
      Console.WriteLine("The number is odd");
}
```

The above IF-Else code can be re-written in a compact, one-line statement using the ternary IF syntax.

```
Console.WriteLine("Please enter a number");
int number = int.Parse(Console.ReadLine());
Console.WriteLine((number % 2 == 0)
      ? "The number is even" : "The number is odd");
```

SWITCH statement

The Switch statement also provides conditional / branching logic. A variable's value is compared to a list of pre-defined values. When a match is found that statement beneath the matching value are executed. It is IMPORTANT to include a BREAK statement at the end of the block where you want the execution to stop, otherwise the program will continue and execute the next case statement block.

There is also a DEFAULT case which gets executed if the variable's value does not match any of the preceding case values.

```
Console.WriteLine("Please enter a number from 1 to 3");
        int number = int.Parse(Console.ReadLine());
        switch(number) {
            case 1:
                Console.WriteLine("You entered 1");
                break;
            case 2:
                Console.WriteLine("You entered 2");
                break;
            case 3:
                Console.WriteLine("You entered 3");
                break;
            default:
                Console.WriteLine("Your number is not in range");
                break;
        }
```

Here is another example where the BREAK statement is deliberately left out of the first two cases, to achieve the result that is required.

```
Console.WriteLine("Please enter a number from 1 to 3");
int number = int.Parse(Console.ReadLine());
switch(number) {
    case 1:
    case 2:
    case 3:
        Console.WriteLine("You entered 1, 2 or 3");
        break;
    default:
        Console.WriteLine("Your number is not in range");
        break;
}
```